Lab 3 Enumeration

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Lab 3-01: Perform NetBIOS Enumeration using Windows Command-Line Utilities

Scenario:

FinanceSafe Inc., a financial services organization, has brought in SecureDES Consultants to conduct a thorough security evaluation of its internal and external network infrastructure. The primary goal is to assess the risk of unauthorized access by identifying sensitive data and weaknesses across the company's Windows-based systems. FinanceSafe's network spans multiple subnets and hosts servers running customer portals, employee management systems, and financial databases. A major concern is potential data leakage through NetBIOS services, which many systems have enabled for file sharing and remote administration purposes.

Solution:

As a Certified Ethical Hacker or penetration tester, the objective is to perform NetBIOS enumeration using native Windows command-line utilities to extract critical internal network details. By leveraging tools such as `nbtstat` and `net use`, the tester can probe the network to uncover active hosts, NetBIOS name tables, shared resources, and user account data that may be vulnerable to exploitation. The intelligence gathered here forms the groundwork for deeper penetration testing and security hardening.

`nbtstat` is a built-in Windows utility useful for diagnosing NetBIOS name resolution issues. It uses case-sensitive switches to retrieve and manage cached NetBIOS data, including NetBT protocol statistics, name tables for both local and remote systems, and the NetBIOS name cache.

> **Note:** In this lab, the target IP address used is **192.168.56.106** (Windows Server 2025). Do not use this IP directly — substitute your own virtual machine's IP address when performing this lab. also we can use NETBIOS ENUMERATOR

Phase 1: Launching the Command Prompt

Start both your **Windows 10** and **Windows Server 2025** virtual machines. Switch to **Windows 10**, use the Windows Search bar on the Desktop, search for **cmd**, and click **Open**.

Phase 2: Querying the Remote NetBIOS Name Table

```
nbtstat -a 192.168.29.107
```

Explanation: The `-a` flag retrieves the NetBIOS name table of a specified remote machine. The result displays the registered NetBIOS names associated with the Windows Server 2025 virtual machine.

```
Wi-Fi 2:
Node IpAddress: [192.168.29.157] Scope Id: []

           NetBIOS Remote Machine Name Table

       Name               Type         Status
    ---------------------------------------------
    VEDANT         <00>  UNIQUE      Registered
    VEDANT         <20>  UNIQUE      Registered
    WORKGROUP      <00>  GROUP       Registered
```

Exception / When to use this: Use this when you need to enumerate the NetBIOS names of a specific remote host to identify machine roles (workstation, domain controller, file server) without requiring a full session or authentication. Showed only my connections in this

Phase 3: Viewing the Local NetBIOS Cache

```
nbtstat -c
```

Explanation: The `-c` switch displays the contents of the local NetBIOS name cache, including resolved remote names and their corresponding IP addresses. This data is retrievable without establishing a null session or any form of authenticated connection.

```
Ethernet:
Node IpAddress: [192.168.29.82] Scope Id: []

                    NetBIOS Remote Cache Name Table

        Name             Type       Host Address    Life [sec]
    ---------------------------------------------------------
    VEDANT         <20>  UNIQUE         192.168.29.107      545

Bluetooth Network Connection:
Node IpAddress: [0.0.0.0] Scope Id: []
```

Exception / When to use this: Use this when you want to review recently resolved NetBIOS entries on the local machine — useful for tracing past communication patterns or confirming earlier enumeration results without generating new network traffic.

Phase 4: Listing Active Network Connections and Shares

```
net use
```

Explanation: The `net use` command outputs information about current network connections, including shared drives, folders, and their connection statuses. This reveals which resources the local machine is currently connected to or has mapped.

```
C:\Users\uniqu>net use
New connections will be remembered.

There are no entries in the list.


C:\Users\uniqu>net view \\192.168.29.107
System error 53 has occurred.

The network path was not found.
```

Exception / When to use this: Use this after initial NetBIOS discovery to identify existing or previously mapped connections to remote shares. Attackers can leverage

this information to read or write to remote systems or initiate Denial-of-Service (DoS) conditions if shares are improperly secured.

Lab 3-02: Perform SNMP Enumeration using SnmpWalk

Scenario:

TechCore Systems, a technology solutions provider, has engaged SecureNet Assessments to conduct a comprehensive security evaluation of its network infrastructure. TechCore relies heavily on routers, switches, and hubs, many of which run the Simple Network Management Protocol (SNMP) for monitoring and management. Concerns have been raised that improperly configured SNMP settings could expose sensitive network information to unauthorized parties.

Solution:

As a Certified Ethical Hacker, the goal is to use SnmpWalk to collect essential details about TechCore's network. This includes identifying active hosts, user accounts, ARP and routing tables, device configurations, and traffic statistics by interacting with SNMP agents running on network devices. The information gathered will help identify vulnerabilities such as weak community strings and excessive device data exposure.

SnmpWalk is a command-line utility that traverses SNMP nodes and retrieves a collection of variables from the target network. When sent to the root node, it collects data from all sub-nodes, including switches and routers.

> **Note:** In this lab, the target IP address used is **192.168.56.107** (Windows Server 2019 core). Use your own VM's IP when performing this lab.

Phase 1: Setting Up the Environment

Power on both **Windows Server 2025** and **ParrotOS** virtual machines. Switch to **ParrotOS**, open the **Terminal**, and elevate to root:

```
sudo su
```

When prompted, enter your root password.

Phase 2: Installing SnmpWalk

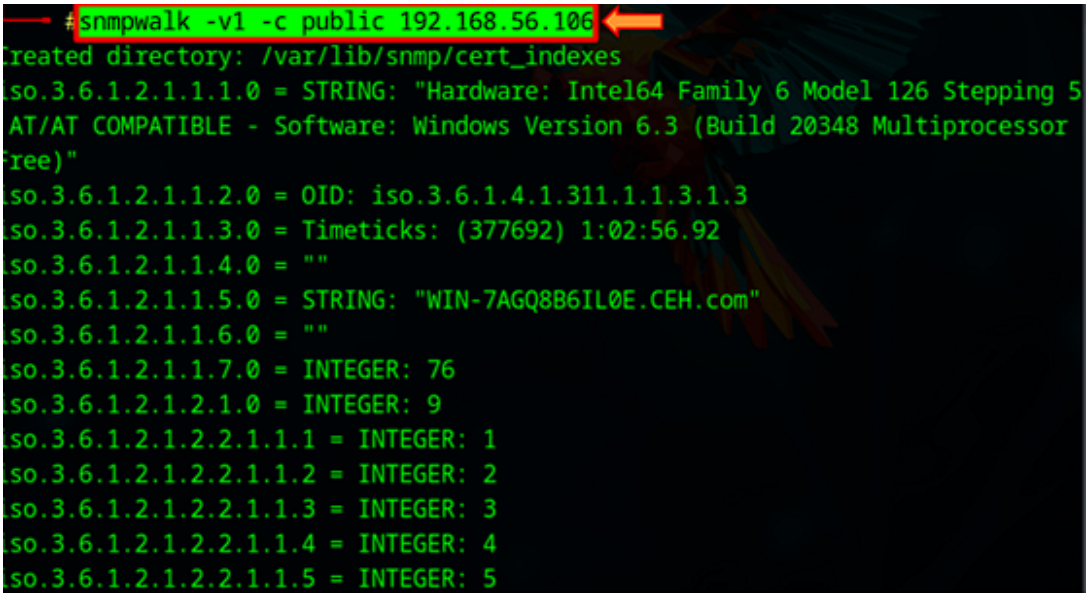
```
apt-get install snmp
```

Explanation: This installs the SNMP utilities package on ParrotOS, including the `snmpwalk` binary needed for enumeration.

Phase 3: SNMP v1 Enumeration

```
snmpwalk -v1 -c public 192.168.56.106
```

Explanation: The `-v` flag sets the SNMP version (1, 2c, or 3), and `-c` specifies the community string (here, `public`). The output lists all OIDs, variable bindings, and related data exposed by the SNMP agent.

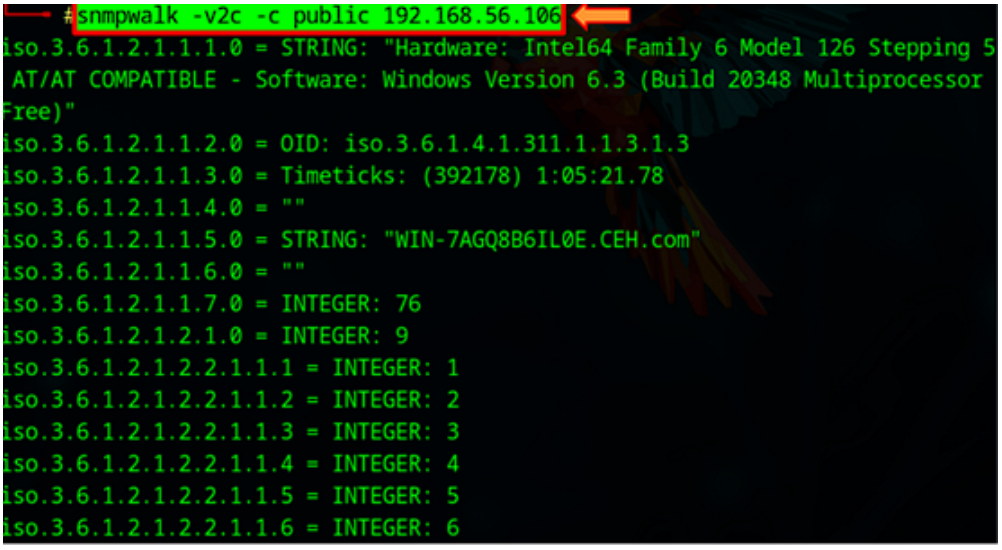


Exception / When to use this: Use SNMPv1 enumeration when targeting legacy devices that only support the older protocol version. Since SNMPv1 uses cleartext community strings, it is the most vulnerable version and the most information-rich for an attacker with network access.

Phase 4: SNMP v2c Enumeration

```
snmpwalk -v2c -c public 192.168.29.107
```

Explanation: This queries the SNMP agent using version 2c, which supports bulk retrieval and is more efficient than v1. The result includes data transmitted from the SNMP agent to the server — covering server details, user credentials, and configuration parameters.



Exception / When to use this: Use this when the target device supports SNMPv2c. It is preferred over v1 for bulk data retrieval but still uses plaintext community strings, making it equally susceptible to sniffing.

Lab 3-03: Perform LDAP Enumeration using Active Directory Explorer (AD Explorer)

Scenario:

GlobalTech Solutions, a multinational organization, has contracted SecureLink Assessments to evaluate the security of its internal directory services. The company uses Active Directory to manage user accounts, devices, and resources across its global offices. The concern is that misconfigured LDAP settings could expose sensitive directory data, making it easier for attackers to map the network and launch targeted attacks.

Solution:

As a Certified Ethical Hacker, the task is to perform LDAP enumeration using Active Directory Explorer (AD Explorer). The objective is to connect to the LDAP server and extract details such as usernames, email addresses, departmental data, server names, and group memberships to identify potential misconfigurations including overly permissive access and weak authentication policies.

AD Explorer is a powerful Active Directory viewer and editor that allows effortless navigation of an AD database, inspection of object properties and attributes, permission management, schema exploration, and execution of advanced saved searches.

> **Note:** The target IP used in this lab is **192.168.56.107** (Windows Server 2019). Substitute your own VM's IP when performing this lab.

Phase 1: Downloading AD Explorer

Start your **Windows 11** and **Windows Server 2025** VMs. On **Windows 11**, open a browser and navigate to:

https://learn.microsoft.com/en-us/sysinternals/downloads/adexplorer

Click **Download AdExplorer** to obtain the executable.

Phase 2: Launching AD Explorer

Go to the **Downloads** folder, unzip the **ADExplorer** archive, and double-click **ADExplorer.exe**. Accept the license agreement by clicking **Agree**.

Phase 3: Connecting to the LDAP Server

In the **Connect to Active Directory** window, enter the **IP address** of Windows Server 2025 in the Connect field, along with the server's **username** and **password**, then click **OK**.

Explanation: This establishes a connection to the LDAP server on TCP port 389, opening access to the directory's hierarchical structure.

Phase 4: Exploring the Directory Structure

Once connected, AD Explorer displays the Active Directory structure in the left pane. Click the **+ icon** to expand **DC=CEH, DC=com**, then expand **CN=Users** to browse domain user objects.

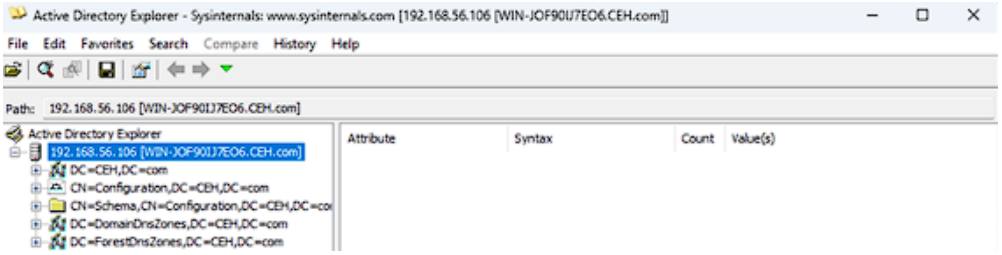
Exception / When to use this: Use this phase to map the organizational structure of the directory — identifying OUs, group memberships, and privileged accounts.

This data is essential for privilege escalation planning and targeted attack surface mapping.

Phase 5: Inspecting and Modifying User Attributes

Click any **username** (e.g., John) in the left pane. In the right pane, right-click the **displayName** attribute and select **Modify...** from the context menu.

In the **Modify Attribute** window, select the username under **Value**, then click **Modify...**. In the **Edit Value** pop-up, update the value in the **Value data** field and click **OK** to save.



Exception / When to use this: Demonstrating write access to directory attributes highlights the risk of overly permissive LDAP configurations. In a real engagement, unauthorized attribute modification could be used for privilege escalation or identity spoofing.

Lab 3-04: Perform NFS Enumeration using RPCScan and SuperEnum

Scenario:

CodeStream Innovations, a software development company, has engaged SecureProbe Analysts to assess its internal file-sharing infrastructure. The company's NFS server stores and manages sensitive assets including proprietary source code, project documentation, and team resources. Concerns exist that misconfigured NFS settings may expose shared directories and client information, creating potential entry points for attackers.

Solution:

As a Certified Ethical Hacker or penetration tester, the task is to perform NFS enumeration using RPCScan and SuperEnum. RPCScan interacts with RPC services

to identify NFS share misconfigurations, listing RPC services, mountpoints, and accessible NFS folders — including recursive enumeration of shares. SuperEnum provides a script that enumerates all open ports, including the NFS port (2049).

> **Note:** The target IP used in this lab is **192.168.56.106** (Windows Server 2025). Use your own VM's IP when running this lab.

Phase 1: Installing NFS on Windows Server 2025

Switch to **Windows Server 2025**, open **Server Manager**, and click **Add roles and features**. Proceed through the wizard, expand **File and Storage Services**, then select the **Server for NFS** checkbox under **File and iSCSI Services**. Confirm by clicking **Add Features**, then click **Next** and **Install**. Close the wizard once installation completes.

Phase 2: Confirming NFS is Active

Switch to **ParrotOS** and run:

```
nmap -p 2049 [Target IP Address]
```

Explanation: The `-p` flag restricts the scan to port 2049. If the NFS service is running, the result will show port 2049 as open.

Phase 3: Installing and Running SuperEnum

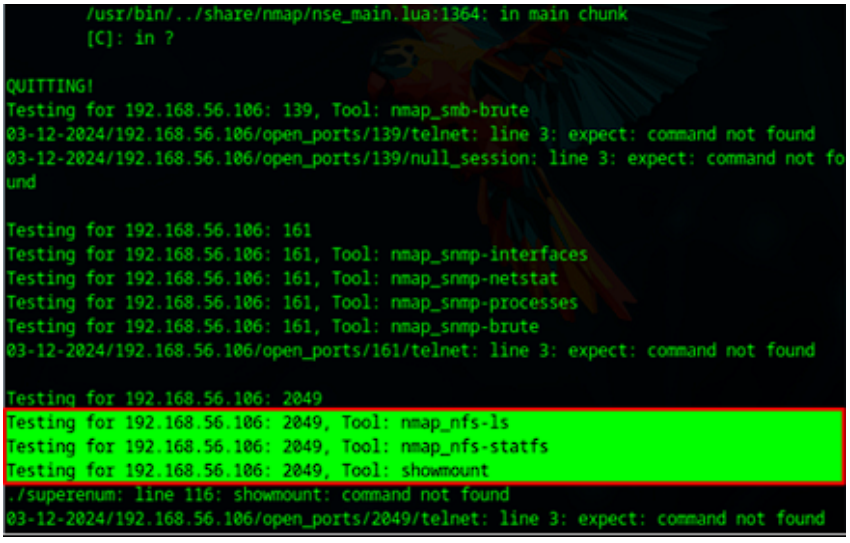
```
git clone https://github.com/p4pentest/SuperEnum.git
cd SuperEnum
echo "192.168.56.106" >> Target.txt
./superenum
```

If a permission error occurs on `./superenum`, grant execute access first:

```
chmod +x superenum
./superenum
```

When prompted, type **Target.txt** for the IP list filename and press **Enter**.

Explanation: SuperEnum scans the target to identify all open ports and services. Once complete, scroll through the output — port 2049 should appear as open, confirming NFS is active.



Exception / When to use this: Use SuperEnum for a broad service sweep across one or more targets when you want to enumerate NFS alongside other services in a single automated pass.

Phase 4: Installing and Running RPCScan

```
cd ..
git clone https://github.com/hegusung/RPCScan.git
cd RPCScan
python3 rpc-scan.py [Target IP address] --rpc
```

Explanation: The `--rpc` flag instructs RPCScan to list portmapper (RPC) services. The result confirms port 2049 is open with NFS running on it and displays available mountpoints and RPC service details.

Exception / When to use this: Use RPCScan when you need a targeted, in-depth view of RPC services and NFS shares specifically. It provides more granular RPC-level detail than a generic port scan, making it useful for identifying specific share misconfigurations that could allow spoofed IP access.

Lab 3-05: Perform DNS Enumeration using Zone Transfer

Scenario:

MarketEdge Solutions, a digital marketing agency, has hired CyberPro Assessments to evaluate its external network security posture. The company manages web assets including client portals, content delivery servers, and DNS infrastructure for its global operations. The concern is that DNS server misconfigurations may expose hostnames, internal IP addresses, and usernames, making the infrastructure easier to map and target.

Solution:

As a Certified Ethical Hacker or penetration tester, the task is to perform DNS enumeration using the Zone Transfer technique. A DNS zone transfer replicates a copy of the zone file from the primary DNS server to a secondary server. If the target DNS server is misconfigured to allow unrestricted zone transfers, it will return a complete record of the domain's structure. Otherwise, it will respond with a transfer failure message.

> **Note:** This lab uses **https://www.certifiedhacker.com/** as the target domain. The name server used is **ns1.bluehost.com**.

Phase 1: Setting Up the Environment

Power on both **Windows 11** and **ParrotOS** VMs. On **ParrotOS**, open the **Terminal** and escalate to root:

```
sudo su
```

Phase 2: Querying DNS Name Servers

```
dig ns [Target Domain]
```

Explanation: The `ns` query type returns the authoritative name servers for the target domain. The results appear in the ANSWER SECTION of the output. On Linux systems, `dig` is the standard tool for querying DNS records including host addresses, name servers, and mail exchanges.

Exception / When to use this: Use this as the first step in DNS reconnaissance to identify which name servers are responsible for the target domain — these become the targets for zone transfer attempts.

Phase 3: Attempting Zone Transfer via dig

```
dig @[NameServer] [Target Domain] axfr
```

Explanation: The `axfr` query type requests a full DNS zone transfer. If the server is not restricted, it will return all DNS records in the zone. A transfer failure message indicates the server has proper controls in place.

Exception / When to use this: Use this against each identified name server to determine whether any permits unrestricted zone transfers. A successful transfer reveals all subdomains, internal hostnames, and IP mappings — critical intelligence for attack planning.

Phase 4: Switching to Windows 11 for nslookup Enumeration

Switch to the **Windows 11** VM. Open **CMD** via the Windows Search bar and run:

```
nslookup
```

Within the nslookup interactive prompt:

```
set querytype=soa
certifiedhacker.com
ls -d [Name Server]
```

Explanation: `set querytype=soa` retrieves the Start of Authority record, revealing the primary name server and responsible mail address for the domain. The `ls -d` command then attempts a zone transfer from the identified name server, requesting all records in the zone.

Exception / When to use this: Use `nslookup` as a Windows-native alternative to `dig` for zone transfer attempts. While `dig` is more feature-rich on Linux, `nslookup` is available on any Windows system without additional tooling.

Lab 3-06: Perform SMTP Enumeration using Nmap

Scenario:

DataLink Solutions, a technology services firm, has engaged ProSecure Assessments to evaluate the security of its email communication infrastructure. The company

relies on an internal SMTP server for departmental and external communications. The concern is that attackers may be able to enumerate valid user accounts or delivery addresses through the SMTP server, enabling phishing campaigns or unauthorized access attempts.

Solution:

As a Certified Ethical Hacker or penetration tester, the objective is to perform SMTP enumeration using Nmap to identify vulnerabilities in the SMTP server configuration. This involves probing the server to discover valid email addresses, usernames, and recipients by leveraging Nmap's scripting engine — specifically scripts such as `smtp-enum-users` — to determine whether the server leaks sensitive account information.

Nmap's scripting engine can be used to enumerate SMTP services and retrieve information about all user accounts on the target SMTP server.

> **Note:** The target IP used is **192.168.56.106** (Windows Server 2025). Use your own VM's IP during this lab.

> **Prerequisite:** SMTP Services must be installed on Windows Server 2025 beforehand. Refer to: https://phoenixnap.com/kb/windows-smtp-server

Phase 1: Setting Up the Environment

Start **Windows Server 2025** and **ParrotOS**. On **ParrotOS**, open the Terminal and switch to root:

```
sudo su
```

Phase 2: Enumerating SMTP Users

```
nmap -p 25 --script=smtp-enum-users [Target IP Address]
```

Explanation: The `-p` flag targets port 25 (SMTP), and `--script` instructs Nmap to run the `smtp-enum-users` script, which probes the server to determine which usernames are valid mail recipients. The output lists all discovered user accounts on the target.

Exception / When to use this: Use this when SMTP (port 25) is found open and the goal is to build a list of valid internal users for subsequent phishing or password

spraying attacks.

Phase 3: Checking for Open Relay

```
nmap -p 25 --script=smtp-open-relay [Target IP Address]
```

Explanation: This script tests whether the SMTP server permits open relaying — the practice of forwarding mail from unauthenticated external sources. Open relays are frequently exploited for spam distribution and phishing campaigns. The output lists relay test results.

Exception / When to use this: Use this to identify servers misconfigured as open relays. An open relay finding is a high-severity vulnerability in any engagement focused on email security hardening.

Phase 4: Listing Available SMTP Commands

```
nmap -p 25 --script=smtp-commands [Target IP Address]
```

Explanation: The `smtp-commands` script retrieves and displays all SMTP commands supported by the target server. This helps map the server's capabilities and identify features that could be abused for further enumeration or exploitation.

Exception / When to use this: Use this to understand the command set available on the server. Supported commands such as `VRFY` and `EXPN` can be used to validate usernames and expand mailing lists — both valuable for user enumeration beyond what `smtp-enum-users` provides.

Lab 3-07: Enumerate Information using Global Network Inventory

Scenario:

FinEdge Solutions, a financial technology company, has partnered with InfoSecure Analysts to conduct a thorough security assessment of its corporate network. The infrastructure includes Windows and Linux machines, file servers, database servers, and Active Directory services. While initial enumeration has yielded useful results, concerns remain that deeper misconfigurations and vulnerabilities may be

overlooked without a more comprehensive audit tool. Complete network visibility has been requested.

Solution:

As a Certified Ethical Hacker or penetration tester, the task is to use Global Network Inventory to perform deep enumeration of the target network's infrastructure. Global Network Inventory is an agentless, zero-deployment audit scanner capable of probing individual machines or ranges by IP or domain as defined in its host file. This exercise helps uncover hidden vulnerabilities such as exposed services, outdated software, and misconfigured settings, while building a complete inventory for prioritizing remediation.

> **Note:** The target IP used is **192.168.56.106** (Windows Server 2025). Substitute your own VM's IP when performing this lab.

Phase 1: Downloading and Installing Global Network Inventory

On **Windows 11**, open a browser and navigate to:

https://magnetosoft.com/products-downloads/

Click **Download Global Network Inventory**, then go to the Downloads folder and double-click **gni_setup.exe**. Proceed through the installation wizard, accepting the license agreement and clicking **Next** through each step. Click **Install**, then **Finish**.

Phase 2: Configuring and Running a Scan

After installation, click **I Agree** on the initial prompt. In the audit scan wizard, under **Audit Scan Mode**, select **Single address scan** and click **Next**. Enter the target machine's **IP address** and click **Next**. Provide your **username** and **password** credentials, click **Next**, then click **Finish** to begin the scan.

The scan process will take a few minutes to complete.

Exception / When to use this: Use single address scanning when targeting a specific machine for a detailed audit. For broader assessments, configure an IP range or import a domain-based host file.

Phase 3: Reviewing Scan Results

Once the scan completes, the **Scan Summary** for the target IP appears. Hover over items under the **Computer** tab to view hardware and system details.

Navigate each tab to retrieve specific data categories:

- **Operating System** — hover over **Windows details** to see full OS information
- **BIOS** — hover over **Default System BIOS** for firmware details
- **NetBIOS** — hover over any listed NetBIOS application for protocol-level data
- **User Groups** — hover over usernames to view group membership details
- **Users** — hover to see login details for each account on the target
- **Services** — hover to view service name, status, and configuration
- **Installed Software** — hover for software name and version data
- **Shares** — hover over shared folders to see their paths and permissions

Additional tabs including **Computer System**, **Processors**, **Main Board**, **Memory**, **SNMP Systems**, and **Hot Fixes** are also available for comprehensive hardware and patch inventory.

Exception / When to use this: Use the **Shares** and **Users** tabs to identify over-exposed resources and accounts for privilege escalation research. The **Hot Fixes** tab is especially valuable for identifying missing patches that map to known CVEs.

Lab 3-08: Perform Enumeration using ShellGPT

Scenario:

While working at a security solutions company, you have been assigned to perform enumeration against a target system. You decide to leverage AI assistance to enhance the process — automating tasks, analyzing patterns, and generating precise commands more efficiently than manual approaches.

Solution:

ShellGPT assists with enumeration by automating command generation, streamlining data analysis, and identifying potential vulnerabilities faster than traditional workflows. The `--shell` flag generates executable shell commands from natural language prompts, while the `--chat` flag maintains session context across multiple related queries.

> **Note:** The commands generated by ShellGPT may vary based on the prompt and environment. Outputs can differ from what is shown here due to the dynamic nature of AI-generated responses. Variations in command syntax, execution, and results are expected.

Phase 1: Prerequisites

Boot the **Parrot Security** VM and log in. Ensure ShellGPT is integrated with an API key before proceeding.

Phase 2: NetBIOS Enumeration

```
sgpt --shell "Perform NetBIOS enumeration on target IP 10.
0.2.15"
```

At the prompt, type **E** and press **Enter** to execute. ShellGPT generates and runs an `nbtstat` or equivalent command against the target.

```
sgpt --shell "Get NetBIOS info for IP 10.0.2.15 and display
the associated names"
```

Explanation: This retrieves NetBIOS name table entries and associated names for the target system.

```
sgpt --shell "Enumerate NetBIOS on target IP 10.0.2.15 with
nmap"
```

Explanation: ShellGPT generates an Nmap-based NetBIOS enumeration command, typically using the `nbstat` NSE script.

Phase 3: SNMP Enumeration

```
sgpt --chat enum --shell "Perform SNMP enumeration on targe
t IP 10.0.2.15 using SnmpWalk and display the result here"
```

Explanation: The `--chat enum` flag maintains context within the session named `enum`. ShellGPT generates an `snmpwalk` command and displays SNMP data from the target.

```
sgpt --chat enum --shell "Perform SNMP enumeration on targe
t IP 10.0.2.15 using nmap and display the result here"
```

```
sgpt --chat enum --shell "Perform SNMP processes on target
IP 10.0.2.15 using nmap and display the result here"
```

Exception / When to use this: Use SNMP enumeration via ShellGPT when you want AI to select the most appropriate tool and flags automatically based on the current session context and findings.

Phase 4: SMTP Enumeration

```
sgpt --chat enum --shell "Perform SMTP enumeration on targe
t IP 10.0.2.15."
```

Explanation: ShellGPT generates an Nmap SMTP script command to enumerate valid users and relay configuration on the target mail server.

Phase 5: DNS Enumeration

```
sgpt --chat enum --shell "Use Nmap to perform DNS Enumerati
on on target domain www.certifiedhacker.com"
```

```
sgpt --chat enum --shell "Use dig command to perform DNS ca
che snooping on target domain www.certifiedhacker.com using
recursive method. Use DNS server IP as 162.241.216.11"
```

```
sgpt --chat enum --shell "Use dig command to perform DNS ca
che snooping on the target domain www.certifiedhacker.com u
sing non-recursive method. Use DNS server IP as 162.241.21
6.11"
```

Explanation: These commands demonstrate DNS cache snooping — probing a DNS resolver to determine whether specific records are cached, revealing which domains have been recently queried by internal users.

Phase 6: IPsec and SMB Enumeration

```
sgpt --shell "Perform IPsec enumeration on target IP 10.0.
2.15 with Nmap. do not use ipsec-enum"
```

```
sgpt --shell "Scan the target IP 10.0.2.15 for the port usi
ng SMB with Nmap"
```

Explanation: ShellGPT generates appropriate Nmap NSE scripts for both IPsec and SMB enumeration without requiring the user to recall specific script names.

Phase 7: Custom Automation Script

```
sgpt --chat enum --shell "Develop and execute a script whic
h will automate various network enumeration tasks on target
IP range 10.0.2.0/24"
```

Explanation: ShellGPT generates and executes a multi-step enumeration script targeting the entire /24 range — combining host discovery, port scanning, service detection, and protocol-specific enumeration in one pass.

Phase 8: LDAP and FTP Enumeration

```
sgpt --shell "Use nmap script to perform ldap-brute-force o
n IP 10.0.2.15"
```

```
sgpt --shell "Use Nmap to perform FTP Enumeration on www.ce
rtifiedhacker.com"
```

Exception / When to use this: ShellGPT-driven enumeration is most effective during broad assessments where speed matters and the tester wants AI to adaptively select tools based on context. For precise, reproducible audits, manually constructed commands are preferred.

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